Calum Hughes

CONTACT INFORMATION

Email: calum.hughes77@gmail.com
Website: calum-hughes.github.io

EDUCATION

PhD in Pure Mathematics, University of Manchester, 2022-2026.

- Awarded Dame Kathleen Ollerenshaw studentship, a competitive research grant which funded my studies.
- Designed and managed my own project, finishing projects before conference deadlines.

MMath Masters in Mathematics, University of Sheffield, 2018-2022.

- First class honours. Finished with an average mark of 90%.
- Studied modules relevant to data analysis: Scientific Investigation Skills, Mathematical Modelling of Natural Systems, Statistics, Time Series.
- Awarded T.M. Flett Prize for the best performance in Mathematics in 2020 and 2022.

PROFESSIONAL SKILLS

Programming:

- Programming languages: Python (NumPy, Pandas, Geopandas, MatPlotLib, PyTorch, scikit-learn), SQL (BigQuery), R (tidyverse), MATLAB, Java and Haskell. Markup languages: LaTeX, HTML and Markdown. Profficient in MS office, in particular excelling in Excel.
- Completed online courses (on Kaggle) in Machine Learning, Geospatial Analysis, SQL, Time Series, Deep Learning and Neural Networks which have helped me apply these skills to various real-world data analysis problems. See the certificates here.
- Utilised the practise of test-driven development throughout my work.

Problem Solving:

- Developed novel mathematical frameworks to solve specific problems that have been used and cited in other published research papers. These solutions have been published in three papers across top-tier mathematics journals.

Communication:

- Gave over 30 research talks at international conferences and seminars to a variety of different audiences.
- Presented my findings to experts in audiences of over 200 people.
- Awarded best presentation at the *Mathematics Students' Research Conference* 2023.
- Taught Python tutorials as a Graduate Teaching Assistant to a class of 80 first year undergraduates, helping them understand the basics of Python and lead a variety of problems classes with numbers of students ranging from 30 to 200. As well as explaining complex concepts in a clear and structured way, I had to distil the skill of problem solving in these students.

Collaboration:

- Went on invited collaborative research visits to Carnegie Mellon University (USA), Masaryk University (Czechia) and the University of Birmingham (UK). This included doing some interdisciplinary research with computer science departments.

Organisation:

- Organised seminars and conferences as part of a committee and personally hosted international guests.
- Secured over £1500 competitive research funding for the organisation of a conference.

Technical Projects

- **Machine Learning for Deforestation Analysis** (Python)
  - Used several machine learning methods from Scikit-Learn on Landsat satellite data from rainforests in Mozambique to estimate forest biophysical parameters, using RMSE to evaluate the best fit.
  - Applied the final model to Landsat satellite data from the Amazon Rainforest to estimate deforestation patterns there.
- **Spatial pattern formation simulator** (Python)
  - Implemented numerical simulations of biological pattern formation, producing visualisations and animations of system evolution.
- **Dynamics of linear operators** (Python)
  - Using Numpy, Pandas and MatPlotLib, I investigated the long-term behaviour of linear operators and used this data to make novel conjectures about the pure mathematical phenomena. This lead to new and interesting results in the area of dynamical systems.
  - From my own project proposal for this, I was awarded an Undergraduate research bursary of £2400 from the London Mathematical Society in 2021. This was a competitive UK-wide place to research a 2-month project and required careful project planning.
- **How to win the lottery** (Python)
  - Designed and developed an algorithm to generate 551 lottery tickets guaranteeing a 3-number match with the UK national lottery.
  - Verified correctness through 1,000,000 simulations, as well as a formal proof.
  - Gave an outreach talk explaining this to undergraduates.

Experience

2025-2026 – **Theory and Applications of Categories**, Academic Reviewer

- Employed because of my expertise in the topics related to the research papers I was reviewing.
- This role requires you to be critical of every detail given in a paper; it requires extreme attention to detail.

2018-2022 – **Glyndebourne Opera House**, Assistant manager for the Orchestra pit

- Worked with orchestras and conductors to design bespoke layouts that worked within the space.
- Managed a team to perform our duties in time-sensitive shifts, requiring excellent time-management, scheduling and leadership skills.

2018-2021 – **Blue Moon Cafe**, Shift Supervisor
2017-2018 – **Sainsbury's**, Store baker

Other Awards

- **Undergraduate Research Internship** from the University of Sheffield School of Mathematics and Statistics, 2020. This was a competitive position and ran for 3 months.
- **SURE Research Grant** of £2000 in 2020 at the University of Sheffield for designing a project entitled "Machine learning and Baroque composition".
- **Best Team Submission** in the School of Mathematics and Statistics annual team Mathematics challenge, 2018.

Additional Interests

- I am interested in gardening, and held many roles in the University of Sheffield's Allotment Society, including President, Treasurer and Finance and Development officer.
- I am a cyclist, and recently cycled from Manchester to Bordeaux!

References available on request

Calum Hughes
26 Withdean Crescent
Brighton
United Kingdom
calum-hughes.github.io
calum.hughes77@gmail.com

Dear
Hiring team,

June 22, 2026

Subject: letter of interest for working at YRGEO

I am writing to register my interest for working at YRGEO, and to enquire whether there are any jobs available which might suit my skill set.

I have just finished a PhD in Pure Mathematics (Logic) at the University of Manchester. My research lies at the intersection of mathematical logic and theoretical computer science, and has resulted in three accepted papers and presentations at international conferences in mathematics and computer science. This work has strengthened my ability to understand complex systems, work with abstract models, and analyse structured data.

Alongside my academic research, I have developed strong programming skills in Python, SQL and R, including experience with numerical computation, statistical analysis, and machine learning tools. I recently completed a project which utilised machine learning methods to estimate forest biophysical parameters. After obtaining datasets for rainforests in Mozambique (for training/ validation data) and the Amazon rainforest (for prediction).I then applied the final model to Landsat satellite data from the Amazon Rainforest to estimate deforestation patterns there. This data was then visualised using GeoPandas.

In fact, I have a particular interest in Geospatial data analysis and have been learning to use software such as QGIS. I recently cycled from Manchester to Bordeaux; before doing this I needed to plan the route taking into consideration cycle paths, quiet roads and campsites as well as places I wanted to visit along the way. To do this, I utilised both QGIS and GeoPandas on python to process and visualise the data. The cycling trip was a huge success!

During my PhD, I wanted to build my communication skills. I gave over 30 research talks to seminars and international audiences (to audiences of 200+, sometimes) of a variety of levels of expertise. Now, I am confident in presenting my knowledge and analyses effectively in a way that meets the needs of the audience.

Teamwork and collaboration are skills that I have exhibited both through my collaborative projects with other researchers, and through my role as part of a committee for organising conferences. I volunteered as part of the committee for the *Mathematics Students' Research Conference* 2023; together we organised a conference for 65+ people and secured £1500 funding from the Heilbronn institute to support our conference.

I work in a careful, structured, and independent manner, developed through long-term research projects requiring sustained focus and attention to detail. I am comfortable working under responsibility and in environments where precision and reliability are essential.

I would welcome the opportunity to contribute my background in problem-solving mathematical modelling, programming, and communication to your team, and to further develop my applied skills.

Your Sincerely,

Calum Hughes